

Parameter	Single latent									Dual latent					
	Held-out			No terms			With terms			No terms			With terms		
	S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3
Traces per 100 m ²	3.75	3.93	3.93	3.50	3.75	3.50	3.50	3.88	3.50	3.50	3.50	3.50	3.50	3.50	3.50
Median length (m)	5.8	5.7	5.7	5.7	5.3	5.5	5.8	5.3	5.6	5.4	5.3	5.2	5.5	5.5	5.1
90th pct. length (m)	13.9	12.4	12.4	10.5	10.1	9.9	11.3	11.5	10.6	10.5	11.1	10.4	11.8	12.1	11.4
Intensity P21 (m/m ²)	0.269	0.265	0.265	0.221	0.221	0.212	0.222	0.244	0.222	0.201	0.217	0.207	0.214	0.223	0.206
Intersections per 100 m ²	0.50	0.41	0.41	0.75	0.62	0.50	0.75	1.00	0.75	0.50	0.57	0.50	0.75	0.75	0.65
Connections per trace	0.73	0.68	0.68	0.78	0.78	0.69	0.82	1.09	0.91	0.69	0.77	0.67	0.90	1.00	0.87
Spacing, nearest (m)	3.1	3.0	3.0	4.0	4.0	4.1	4.0	3.9	3.8	4.1	4.1	4.2	4.1	4.1	4.0
C1 (~125°) share (%)	36	34	34	23	22	23	36	35	35	20	18	18	34	34	36
C2 (~56°) share (%)	32	31	32	20	24	24	25	29	27	26	30	29	28	28	28
C3 (~27°) share (%)	19	19	18	24	23	22	17	17	19	24	22	27	19	20	18
C4 (~89°) share (%)	10	10	10	11	9	10	14	11	13	11	12	10	13	10	13
C1 direction (°)	125	125	124	126	126	125	127	124	124	126	125	125	126	125	124
C2 direction (°)	56	57	56	56	56	56	56	55	56	57	56	57	56	56	56
C1 spread (°)	10.1	11.0	10.9	12.2	12.9	12.9	11.9	12.3	12.4	12.5	13.5	13.3	11.6	12.0	12.2
C2 spread (°)	9.3	9.8	10.3	10.3	10.7	10.3	10.3	10.5	10.7	10.1	10.6	10.4	10.0	10.4	10.6

Dual latent with terms = the full EVAE (green). Bold = closest to held-out in that test (ties: all bold).
Terms = geology-informed terms (geology, trace-direction and cluster terms). C1-C4 = trace-direction clusters. Same seeds in every column
S1 fill a gap (20 test boxes); S2 new bench level (50); S3 new part of the wall (50).